Fixed-threshold transfer and equal-review ranking in breast ultrasound segmentation quality control

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Running title: Threshold transfer and review budgets

Abstract

Objectives To distinguish transfer of fixed numerical review thresholds from ranking at a fixed review count for breast ultrasound segmentation quality control.

Methods Frozen Segment Anything Model predictions used reference-derived, known-location boxes. Confidence and prompt-disagreement thresholds targeting 20 % review in 213 development patients were transferred to 851 internal and 252 external patients. Low quality was reference Dice <0.70. Equal-review ranking and a post hoc eight-budget analysis assessed policy dependence.

Results Confidence-based detection was 71/93 internally and 10/16 externally: −13.84 percentage points (95 % confidence interval −38.85 to 7.45). Review fractions were 20.45 % and 13.89 %, respectively. Externally, disagreement detected 13/16 low-quality masks at its transferred cutoff versus 10/16 for confidence; these counts were unchanged in post hoc readouts at matched 33 or 35 reviews. At 50 reviews per method, the corresponding counts were 13/16 and 14/16. Across targets from 5 % to 50 %, neither score consistently detected more low-quality masks.

Conclusions Fixed-threshold transfer and equal-review ranking answer different deployment questions. Observed score comparisons depended on the policy and budget; external detection remained imprecise. Reporting detection, review fraction and misses together makes the operating rule explicit.

Keywords domain shift; external testing; prompt perturbation; selective prediction; vision foundation models

Introduction

The Segment Anything Model (SAM) can delineate an object from a spatial prompt, such as a box around a lesion [1]. Medical adaptations of SAM have evaluated box-prompted segmentation across several modalities, including breast ultrasound [2]. When a segmentation is used to support a measurement or other downstream analysis, a separate question is which outputs should be sent for review. A high overall Dice coefficient can coexist with a subset of masks that have poor agreement with a reference contour.

Reverse classification accuracy estimates case-level segmentation quality without a new manual mask [3]. Kinalioglu combined structural prompt responses with confidence and reported failure recovery among the highest-risk 10 % of masks [4]. Loke et al. compared prompt-perturbation uncertainty with predicted IoU, entropy and test-time augmentation across SAM/MedSAM using paired tests and multiplicity control [5]. Building on these scoring and ranking studies, we examine development-calibrated numerical cutoff transfer to internal and external breast ultrasound cohorts. Comparisons at the same actual review counts and joint reporting of detection, workload and misses distinguish cutoff transfer from capacity-constrained ranking.

We evaluated these two operating policies using model-confidence and prompt-disagreement risks on identical initial masks from a frozen SAM. Each score was calibrated to a 20 % development review target in BUS-BRA, then its numerical cutoff was applied unchanged to internal BUS-BRA and external BrEaST patients. The prespecified primary contrast, H1, compared external-minus-internal low-quality detection for confidence-based method A; H2 assessed its review fraction, and H3 compared the two scores on the same low-quality external masks at the locked cutoffs. Equal-review ranking addressed the capacity-constrained question separately. Complete-budget and resampling analyses examined the scope and stability of the comparison, while diagnostic-stratum accounting and prediction contours provided context for workload differences and missed cases.

Materials and methods

Study design and datasets

This retrospective evaluation used anonymized images and manual lesion contours from two public breast ultrasound datasets. Reporting was informed by the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) 2024 update [6]. The patient was the analysis unit,

with one image and one primary reference mask per patient. Figure 1 shows the development, internal test and external test cohorts. Table 1 describes their composition.

BUS-BRA version 1.0 contains 1,875 images from 1,064 patients [7, 8]. We used a patient-level split stratified by benign or malignant lesion status to obtain 213 development and 851 internal test patients (seed 20261105). We selected one eligible image per patient using seed 20261106. The two cohorts had disjoint patient identifiers. We verified image and mask pairs and used the actual pixel dimensions when these differed from metadata. An initial 40-case feasibility run used a subset of development patients. The full development set supplied the formal thresholds.

The BrEaST version 1 archive includes 256 scans, each from a different patient [9, 10]. We matched the official image and main tumor mask by case identifier. Four normal cases lacked a primary lesion mask, leaving 252 external test patients. The primary reference was the provided main tumor contour. Fourteen additional masks were outside that reference definition. The audit found zero exact image matches between the datasets.

[Place Table 1 near here]

Prompted segmentation and risk scores

We used the official SAM ViT-B checkpoint (sam_vit_b_01ec64.pth) with frozen float32 weights and single-mask output [1]. The original source commit, checkpoint hash and software versions are recorded in Table S1. We decoded images to red, green and blue (RGB) channels. BrEaST images were alpha-composited over black before conversion to RGB. This changed the RGB pixel values of eight images relative to discarding the alpha channel. Reference masks were read as binary arrays.

We obtained the initial prompt box from the tight bounding box of the nonempty reference mask, using half-open pixel bounds with upper coordinates one pixel beyond the last foreground index, expanded each horizontal side by 10 % of the width and each vertical side by 10 % of the height, and clipped the result to the image. This is a reference-informed, known-location (oracle-localization) setting, not a test of autonomous lesion localization. We encoded the image once and used the resulting embedding for the initial prediction and four additional predictions (Figure 2). The fixed 10 % context margin was used consistently across datasets.

For each image, let P_0 be the initial binary prediction, P_j its prediction under shifted prompt j, and G the manual reference mask. For binary masks U, V with at least one foreground pixel between them, Dice agreement is

$$D(U,V)=\frac{2|U\cap V|}{|U|+|V|}. \tag{1}$$

Here $|\cdot|$ counts foreground pixels. For consistency scoring, Dice is defined as one when both predicted masks are empty. Method A uses the confidence risk in Eq. (2):

$$r_{\mathrm{A}}=1-\hat{q}. \tag{2}$$

Here $\hat{q}$ denotes the raw predicted intersection over union (IoU) returned by the SAM quality head for the initial mask. For method B, we shifted the initial clipped box left and right by 5 % of its width and up and down by 5 % of its height, clipping each shifted box to the image. The four initial-to-shifted Dice values define the disagreement risk

$$r_{\mathrm{B}}=1-\frac{1}{4}\sum_{j=1}^{4} D(P_0,P_j). \tag{3}$$

All four directions were retained when clipping produced coincident boxes. These deterministic shifts probe local stability relative to box size. Method B adds four mask-decoder passes while reusing the image embedding. Recorded inference times cover the combined pipeline; A-only and incremental B times were not separately benchmarked.

The quality reference for both methods was the initial prediction's Dice against the manual mask, $D(P_0,G)$ in Eq. (1). An empty initial prediction against the nonempty reference has Dice zero. Low quality was defined as reference Dice below 0.70, an operational cutoff for this evaluation. The raw model confidence in Eq. (2) and prediction-consistency score in Eq. (3) remain separate from this reference-quality outcome.

OpenAI Codex assisted with study-design discussions and implementation of research, statistical and plotting code. The pipeline processed public images with frozen SAM weights, and figures were rendered conventionally from recorded data and captured predictions. The human authors retain responsibility for the implementation and calculated outputs. A precise underlying Codex model identifier was not available in the archived study records.

Development threshold selection

For each method, we searched the distinct development risk scores and the option to flag zero cases. A case was flagged if its score met or exceeded the candidate threshold. The selected

threshold gave the review fraction closest to 20 %. Cases tied at the cutoff were flagged together. When two choices were equally close to the target, we selected the one with fewer flagged cases. The exact thresholds were locked before test scoring and applied to both test cohorts. The 20 % target defines a prespecified one-in-five review scenario rather than an empirically measured clinical review capacity.

Outcomes and statistical analysis

The analysis included cases with a valid reference Dice and both finite risk scores. The procedure allowed one retry with the same configuration after a technical failure and recorded persistent failures separately. We report low-quality detection as the number of flagged low-quality masks divided by the number of low-quality masks. Simulated review is the number of flagged cases divided by all scored cases. Misses per 100 patients is 100 times the number of unflagged low-quality masks divided by all scored cases. Rates with a zero denominator were left undefined.

The dated analysis plan specified three contrasts. H1, the primary contrast, compared method A low-quality detection in BrEaST minus BUS-BRA internal testing. H2 used the same cohort order for method A simulated review. H3 assessed whether method B detected a larger fraction of the same low-quality BrEaST cases when the two cutoffs gave equal review counts in development. The paired contrast was B minus A. Differences are expressed in percentage points. We calculated two-sided 95 % confidence intervals (CIs) using Newcombe score method 10 for independent proportions in H1 and H2 [11] and its paired counterpart in H3 [12]. H2, H3 and supplementary intervals are nominal, with no multiplicity adjustment.

We repeated detection analyses with reference Dice cutoffs of 0.50 and 0.80 while retaining both locked risk thresholds. The original protocol additionally specified an equal-review supplementary comparison in BrEaST: each score ranked the external cases and selected the top 50, the count closest to a 20 % review fraction among 252 cases. Risk ties followed the unmodified archived patient-identifier strings in lexicographic order (for example, "1", "10", "100"), rather than numeric or zero-padded image-identifier order. This planned supplementary comparison used a common review count in the observed external cohort; its paired interval is conditional on the observed rankings. It is separate from H3, which compares the development-locked cutoffs.

As post hoc descriptive quality-control measures, we additionally tabulated true-positive (TP), false-positive (FP), true-negative (TN) and false-negative (FN) counts, sensitivity,

specificity, positive predictive value (PPV) and negative predictive value (NPV), with Wilson 95 % intervals (Tables S6 and S7). Low-quality segmentation was the positive condition and a review flag the positive test. These measures assess mask-quality screening, not diagnosis of breast cancer.

We additionally performed a post hoc exploratory stratification by dataset-reported benign/malignant classification, retaining the locked risk thresholds and Dice < 0.70 definition. BUS-BRA labels came from the official Pathology field, linked by image and patient identifiers. BrEaST labels came from Classification, linked by CaseID and verified against the selected filenames; 197 diagnoses were biopsy-confirmed and 55 were confirmed by follow-up. We report stratum counts, review fractions and detection with Wilson 95 % intervals. No stratum-specific recalibration, hypothesis test or interaction test was performed. We also audited all external low-quality cases missed by A, partitioned by the B review flag.

Post hoc policy sensitivity and diagnostic accounting

After the original study, we conducted additional post hoc policy and diagnostic-accounting analyses. These retain the original scores, low-quality definition and H1–H3. We examined development review targets of 5 %, 10 %, 15 %, 20 %, 25 %, 30 %, 40 % and 50 %, calibrating a separate threshold for each method with the same tied-group and nearest-target rules. Selection used development risks and counts, without quality labels. Each numerical threshold was transferred unchanged to both test cohorts. A separate equal-review policy selected the nearest integer to the target fraction of each test cohort, taking fewer cases at an exact half-integer tie.

Post hoc descriptive readouts were taken at 33 and 35 reviews per method, the two external volumes produced by the transferred cutoffs; the 50-review readout repeated the original-protocol supplementary budget. All were extracted from the archived external ranking curves without new inference, calibration, intervals or hypothesis tests. The original contrasts and intervals remain unchanged.

We used 10,000 patient-bootstrap replicates per route (seed 20261106) to separate three sources of variation: test-cohort sampling at the original cutoffs; development resampling and cutoff recalibration with the original test cohorts unchanged; and both together. Cohorts were sampled independently, with paired A/B scores and labels retained within every record. Repeated draws were retained. Equal-review ranking selected exactly 50 external draw records, ordered by descending risk, original identifier string and draw-position index. The

strict original direction pattern required external B-minus-A detection above zero at transferred cutoffs and below zero at equal review; ties were reported separately. Percentile 95 % intervals are descriptive, nominal sensitivity intervals and do not replace the prespecified Newcombe intervals.

We also used symmetric Kitagawa accounting and direct standardization across dataset-reported benign/malignant strata [13]. The external-minus-internal difference was partitioned into a composition-associated term and a within-stratum-rate term. Common weights were the arithmetic mean of the two cohort weights. Detection used the diagnostic shares among low-quality masks, whereas review fractions used shares among all scored patients. We obtained nominal percentile intervals from 5,000 independently sampled cohort-level patient-bootstrap replicates (seed 20261107), recalculating all weights and rates. Undefined stratum denominators remained missing. This accounting describes the observed mixtures and does not estimate causal effects. The two component definitions are given in Eqs. (S1) and (S2) with Table S12.

Computational reproducibility

The original study assessed computational reproducibility by recomputing statistics from archived case scores and by rerunning all 1,316 images in the original computing environment. Its recorded comparisons covered eight numerical fields, quality labels, review decisions, calibration and review-curve counts. For the present revision, we additionally replayed the six A-missed external cases on a CPU using the checksum-identical original checkpoint, exact SAM source and frozen float32 prompt rules. We saved the initial and four shifted predictions for each case and compared all eight numerical fields and decisions with the archive. The two exercises have separate environments and receipts; all cohort analyses continue to use the original scores.

Results

Cohort yield and development calibration

All 213 development, 851 internal test and 252 external test cases were scored. Low-quality masks occurred in 24/213 (11.27 %) development cases, 93/851 (10.93 %) internal cases and 16/252 (6.35 %) external cases. Each method flagged 43/213 (20.19 %) development cases. The locked thresholds were 0.13705134391784668 for A and 0.06972856653991777 for B.

No initial prediction was empty in the 1,316 study images. All 5,264 initial-to-shifted consistency Dice values were strictly positive, establishing that all shifted predictions were also nonempty. Thus the empty-empty Dice convention did not affect any recorded risk score or review decision.

Performance at the locked thresholds

For H1, method A flagged 71 of 93 low-quality masks in BUS-BRA internal testing (76.34 %) and 10 of 16 in BrEaST (62.50 %; Wilson 95 % CI 38.64 to 81.52 %). The external-minus-internal difference was −13.84 percentage points (95 % CI −38.85 to 7.45). The interval includes zero and is compatible with both a substantial decrease and a smaller increase in detection (Tables 2 and 3 and Figure 3).

[Place Tables 2 and 3 near here]

For H2, the method A simulated review fraction was 174/851 (20.45 %) internally and 35/252 (13.89 %) externally, a difference of −6.56 percentage points (nominal 95 % CI −11.25 to −1.10). Among masks with reference Dice of at least 0.70, A flagged 103/758 (13.59 %) internally and 25/236 (10.59 %) externally. The overall rates of missed low-quality masks with A were 2.59 per 100 internal patients and 2.38 per 100 external patients.

For H3, method B flagged 13/16 (81.25 %; Wilson 95 % CI 56.99 to 93.41 %) low-quality BrEaST masks compared with 10/16 (62.50 %) for A. Ten were flagged by both scores, three by B alone and three by neither. The paired difference was 18.75 percentage points (nominal 95 % CI −4.28 to 40.04). B flagged 33/252 (13.10 %) external cases and missed three low-quality masks. In the internal cohort, B flagged 150/851 (17.63 %) cases, detected 68/93 (73.12 %) low-quality masks and missed 25 (Table 2).

Sensitivity and equal-review comparisons

At reference Dice below 0.50, A detected 52/53 internal and 5/7 external low-quality masks, and B detected 43/53 and 5/7, respectively. At Dice below 0.80, the corresponding counts were 109/183 and 21/50 for A and 95/183 and 21/50 for B. With each alternative definition, A's external detection point estimate was lower than its internal estimate, and both methods had the same external detection count (Tables S2 and S3 and Figure S1).

At 50/252 reviews (19.84 %) per method in BrEaST, A detected 14/16 (87.50 %) and B detected 13/16 (81.25 %) low-quality masks. The paired B-minus-A difference was −6.25 percentage points (nominal 95 % CI −31.61 to 19.57). The order of the two detection point

estimates differed between the locked-threshold and equal-review settings (Figure 4 and Table S4). Figure S2 shows the paired detections and misses in both settings.

At matched counts of either 33 or 35 reviews, A detected 10/16 low-quality masks and B detected 13/16, retaining the three-case difference observed at transferred cutoffs (Table S4 and Figure 4D). Increasing the common budget from 35 to 50 added four low-quality detections for A and none for B. The ordering therefore also varied with budget under the same ranking rule. These post hoc readouts describe this cohort without establishing population-level superiority.

Reproducibility checks

The archived same-environment raw-image rerun scored all 1,316 cases. The maximum absolute difference was zero for each of the eight numerical fields, and all quality and review decisions matched (Table S5). Statistical replay reproduced the thresholds, primary and supplementary results, and count-level review curves. These checks concern computational reproducibility of the recorded pipeline.

Additional descriptive quality-control measures

The post hoc external quality-control counts were TP=10, FP=25, TN=211 and FN=6 for A, and TP=13, FP=20, TN=216 and FN=3 for B. Specificity was 89.41 % for A and 91.53 % for B; PPV was 28.57 % and 39.39 %, and NPV was 97.24 % and 98.63 %, respectively. The complete counts and Wilson intervals are provided in Tables S6 and S7.

Exploratory diagnostic strata

Internal benign- and malignant-labelled groups contained 577 and 274 patients, including 53 and 40 low-quality masks. A detected 51/53 (96.23 %; Wilson 95 % CI 87.25 to 98.96 %) and 20/40 (50.00 %; 35.20 to 64.80 %), while B detected 41/53 (77.36 %) and 27/40 (67.50 %), respectively. Externally, each diagnostic stratum contained eight low-quality masks: A detected 5/8 in each group, and B detected 7/8 benign-labelled and 6/8 malignant-labelled cases. Review fractions and intervals are reported in Table S8. All stratum counts recombine exactly to the overall results.

External missed-case audit

The complete set of six external low-quality cases missed by A comprised three also missed by B (cases 089, 175 and 238) and three detected by B (024, 105 and 222). In the shared-miss group, mean prompt-consistency Dice was 0.9590, 0.9799 and 0.9942, respectively, despite

reference Dice of 0.3191, 0.0293 and 0.6813. Table S9 reports all six cases, their archived scores and flags. This is an outcome-defined descriptive case audit, not an additional estimate of population performance.

Review-policy and calibration sensitivity

Across the complete budget grid, external B-minus-A detection under transferred cutoffs was positive at targets of 5–20 %, negative at 25 %, 30 % and 50 %, and zero at 40 %. Under equal-review ranking, it was positive at 5–15 %, negative at 20 %, and zero at 25–50 % (Table S10 and Figure S3). Neither score consistently detected more low-quality masks across these policies and budgets.

The strict 20 %-target direction pattern occurred in 39.61 % of fixed-cutoff test-sampling replicates, 90.40 % of development-only calibration replicates and 30.94 % of full development-and-test replicates. At least one comparison was tied in 28.08 % of full replicates. Full-resampling percentile intervals were [−39.03, +11.71] percentage points for A detection and [−12.70, −1.55] for its review fraction, in external-minus-internal order (Table S11 and Figure S4). These frequencies describe empirical resampling, not p-values or population probabilities. At the original fixed cutoffs, the absence of an observed A-only low-quality detection structurally constrains the test-sampling bootstrap difference to be nonnegative; its [0.00, +40.00] percentile interval cannot exclude a population A advantage. The original H3 Newcombe interval remains unchanged.

Diagnostic composition and standardized differences

For A, the −6.56-point external-minus-internal review difference comprised a +0.64-point composition-associated term and a −7.20-point within-stratum-rate term. Its detection difference had corresponding terms of −1.62 and −12.23 points. For B, the review terms were +0.79 and −5.32 points, and the detection terms were −0.78 and +8.91 points (Table S12 and Figure S5). The within-stratum term equals the difference after standardization to common diagnostic weights. Detection components were defined in 4,999/5,000 replicates; one replicate had no external benign low-quality case. Overall differences and all review components had 5,000 valid replicates.

Spatial inspection from targeted CPU replay

The six-case CPU replay saved 30 nonempty prediction masks. All 12 A/B review decisions and six quality labels matched the original records. Initial reference Dice differed by at most

9.61 × 10−5; the largest difference across all 48 numerical comparisons was 0.00143543 for left-shift consistency Dice in case105 (Table S13). Figure 5 shows the three shared misses, with all six cases in Figure S6. In case175, the predicted region lay along the upper image edge and had little overlap with the released reference (replayed Dice 0.0293), while both risks remained below their review cutoffs. These are newly captured CPU masks; the original six CUDA arrays were not archived, so pixel identity could not be tested.

Discussion

Decision rules and deployment workload

A numerical cutoff preserves a development decision threshold, whereas ranking at a fixed count allocates a specified number of reviews within a cohort or batch. In BrEaST, the transferred cutoffs selected 35 patients with A and 33 with B. At matched counts of either 33 or 35, B still detected 13/16 low-quality masks and A detected 10/16; the count difference at the cutoffs therefore was not explained solely by their unequal review volumes. At 50 reviews each, A detected 14/16 and B detected 13/16. The ordering also changed as the budget varied under the same equal-review ranking rule. Reporting the numerical cutoff, actual review count, detections and misses together makes both the operating rule and its budget explicit, without attributing the ordering change to policy alone.

Method A selected approximately one fifth of the internal cohort and one seventh of the external cohort for review. Its external detection contrast was less precise: workload was estimated from all 252 external patients, whereas detection used the 16 low-quality masks. The review fraction combines the prevalence of low-quality masks with the flagged fractions in low-quality and acceptable groups. BrEaST had both a smaller low-quality fraction and a smaller flagged fraction among acceptable masks. Misses per 100 patients depend on that prevalence too, explaining why they can remain similar while detection point estimates differ.

The diagnostic-stratum analysis gives a more specific description of this workload difference. For A, the positive composition-associated term and negative within-stratum-rate term act in opposite directions; the latter is also the common-weight standardized difference. Thus the two-stratum accounting locates the observed difference in rates within the diagnostic groups, without identifying its cause. For detection, these weights describe the low-quality subset, not all patients. Dataset differences in acquisition, devices and reference annotation remain combined with the observed diagnostic mixtures.

What prompt disagreement adds

Prompt disagreement added three detections at the locked external operating point. Its yield relative to confidence was not uniform across budgets, and the bootstrap characterized the sensitivity of the 20 % direction pattern to calibration and test sampling. The development-only result is conditional on the original test patients and rankings; the full-resampling result also incorporates their empirical variation. Together with the original paired interval, these analyses support a policy-specific interpretation of the observed counts rather than a general ordering of the scores.

Confidence and prompt disagreement capture different properties of the same initial prediction. A uses the model quality estimate, whereas B probes stability under four specified box translations. The replay contours make their shared blind spot visible: a consistently selected region can remain poorly aligned with the reference. Case175 is an inspectable example of this behavior, with a narrow upper-edge prediction and very low reference Dice.

Budget-constrained failure screening is already established [4], as are cross-dataset and prompt-regime comparisons of uncertainty scores [5]. Here, the transferred cutoff leaves workload as an outcome, while equal-count ranking fixes the number selected. Our resampling routes separate calibration and test-sampling uncertainty. The matched-count readouts further identify budget-dependent ordering within this operating example.

Limitations and implications

The findings apply to reference-informed, known-location boxes. Reader-drawn or automatically generated boxes require separate evaluation. The study used one SAM checkpoint, one image per patient and one independent external dataset; overlap with the SA-1B pretraining archive could not be determined [1]. Sixteen external low-quality masks limit detection precision, and empirical resampling does not add independent patients. The new budget, resampling and accounting analyses are post hoc; they define the scope of the operating example rather than confirming a prespecified population-level reversal.

Reference Dice measures agreement with released contours, and the operational cutoff requires separate clinical justification. Simulated review counts do not measure reader time, correction success or downstream measurement accuracy. Future evaluations should predefine the review policy and assess detection and workload jointly with independently supplied prompts and additional external cohorts.

All recorded predictions were nonempty. In future data, simultaneous empty initial and shifted masks could nevertheless receive zero disagreement risk under the stated consistency convention; any explicit empty-mask review rule would be a prospective policy change.

Conclusions

In this known-location breast ultrasound evaluation, transferring development-calibrated numerical cutoffs changed the fraction of cases flagged for review. The relative detection yield depended on the operating policy and budget, with no uniformly better score across the complete grid. Joint reporting of detection, review fraction and missed cases, together with an explicit choice of fixed cutoff or fixed review count, provides a practical basis for interpreting segmentation-quality screening across datasets.

Research ethics. This retrospective secondary analysis used lawfully obtained, publicly released anonymized breast ultrasound images and annotations and involved no participant recruitment or intervention. The authors confirmed that the study met the conditions for exemption from ethics review under Article 32 of the Chinese Measures for Ethical Review of Life Science and Medical Research Involving Human Participants (2023).

Informed consent. No participants were recruited or contacted for this secondary analysis of publicly available anonymized data, and no additional informed consent was obtained.

Author contributions. All authors have accepted responsibility for the entire content of this manuscript and approved its submission. Haoze Lan: research conceptualization and principal study design. Ziyuan Zhu: code review, GitHub repository uploads and manuscript revision.

Use of Large Language Models, AI and Machine Learning Tools. OpenAI Codex assisted with study-design discussions, research and statistical code, plotting scripts, and substantive drafting and editing. Figures were rendered with Python/Matplotlib from released images, captured SAM masks and recorded scores. Editorial drafting used the Scientific Agent Skills workflow [14]. A precise underlying Codex model identifier was not available in the archived study records. The human authors retain responsibility for the work.

Conflict of interest. The authors state no conflict of interest.

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Data availability. The original BUS-BRA version 1.0 images and reference masks are available at https://doi.org/10.5281/zenodo.8231412 and BrEaST version 1 at https://doi.org/10.7937/9WKK-Q141 [8, 10]. Study-specific manifests, case scores, scoring

and statistical code, and reproducibility records are available in the research repository [15]. The original study archive is retained at commit 592566b4c5f39e261c676eb034139a974e9cdfd5; the additional budget, bootstrap, diagnostic-accounting and six-case replay evidence is archived at commit 6a7da8181db6ba3db0bb9fa8bef628f284791b2e. Original datasets and derived materials are distinguished in the data, experiments and reproducibility directories. The checkpoint is not bundled; documented source URLs and its SHA-256 allow identity verification. The supplementary materials include the new tables, actual replay contours and executable analysis references. Matched-count presentation data and analysis scripts are supplied in the current submission directory under Analysis/Policy Clarity, with Breast Ultrasound Quality Control Matched Review Counts.csv provided as supplementary data.

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Tables and table legends

Table 1. Study cohorts

Cohort	Patients, n	Benign, n	Malignant, n	Low-quality masks, n/N (%)
BUS-BRA development	213	145	68	24/213 (11.27 %)
BUS-BRA internal	851	577	274	93/851 (10.93 %)
BrEaST external	252	154	98	16/252 (6.35 %)

One image was selected per patient. Low-quality masks had a reference Dice coefficient below 0.70. n, number of patients; N, cohort size.

Table 2. Test performance at the locked review thresholds

Test cohort	Score	Detection, n/N (%) [Wilson 95 % CI, %]	Simulated review, n/N (%)	Misses, n	Misses per 100 patients
BUS-BRA internal	A	71/93 (76.34 %) [66.77, 83.83]	174/851 (20.45 %)	22	2.59
BUS-BRA internal	B	68/93 (73.12 %) [63.32, 81.08]	150/851 (17.63 %)	25	2.94
BrEaST external	A	10/16 (62.50 %) [38.64, 81.52]	35/252 (13.89 %)	6	2.38
BrEaST external	B	13/16 (81.25 %) [56.99, 93.41]	33/252 (13.10 %)	3	1.19

Detection uses low-quality masks as its denominator. Simulated review and misses per 100 use all scored patients in each cohort. A, model-confidence score; B, prompt-disagreement score; n, number of events; N, relevant denominator. Brackets show Wilson 95 % intervals for detection, in percent.

Table 3. Planned contrasts at the locked review thresholds

Analysis	Contrast	Difference, pp	95 % CI, pp
H1 (primary)	Method A detection, external minus internal	−13.84	[−38.85, +7.45]
H2 (secondary)	Method A simulated review, external minus internal	−6.56	[−11.25, −1.10]
H3 (secondary)	External detection, Method B minus Method A	+18.75	[−4.28, +40.04]

pp, percentage points. CI, confidence interval. H1 and H2 use independent Newcombe intervals. H3 uses a paired interval in the 16 low-quality BrEaST cases. H2 and H3 intervals are nominal.

Figure legends

Figure 1. Cohort construction and fixed-threshold evaluation. (A) BUS-BRA patients are split into development and internal testing; the development cohort selects each review cutoff at a 20 % target. (B) BrEaST contributes 252 patients after exclusion of four normal cases without a primary lesion mask. Each analysis patient contributes one image. Development cutoffs are transferred unchanged to the 1,103 internal and external test patients. Low-quality masks have initial-prediction Dice below 0.70 against the released reference. Counts refer to patients and their single initial masks.

Figure 2. Reference-informed prompting, observed SAM outputs and separate review scores. (A) The released image and reference from BrEaST case001 [10] define the initial box after 10 % expansion per side. Actual archived initial, left-, right-, up- and down-prompt masks are shown in their full image field; white is foreground. Shifts are 5 % of the clipped initial width or height. (B) Frozen SAM ViT-B encodes each image once and uses the shared embedding for five single-mask predictions. (C) Method A is one minus initial-mask predicted IoU; method B is one minus the mean initial-to-shifted mask Dice. Each score is evaluated separately against its locked development cutoff. The reference-derived prompt represents known-location evaluation.

Figure 3. Prespecified patient-level contrasts. Points show differences in percentage points; horizontal bars show original two-sided 95 % Newcombe method 10 confidence intervals. H1 compares Method A low-quality detection in the BrEaST external test cohort with the BUS-BRA internal holdout. H2 compares Method A simulated review fractions between those cohorts. H3 compares low-quality detection for Method B minus Method A among the same 16 external low-quality patients and uses the original paired interval. H2 and H3 intervals are nominal without multiplicity adjustment. Low quality is Dice <0.70. Numerators and denominators are shown for each contrast.

Figure 4. Risk-ranked review curves, original paired contrasts and matched counts. A/B: internal (93 low-quality/851 patients) and external (16/252) curves, ranked by descending risk and lexicographic CaseID. Low quality is Dice <0.70. Blue solid/circle and orange dashed/triangle encodings identify A and B. Filled squares mark locked cutoffs: internal A, 71/93 at 174 reviews; B, 68/93 at 150; external A, 10/16 at 35; B, 13/16 at 33. Open circles mark 50 external reviews per method (A: 14/16; B: 13/16). C: B-minus-A detection differences with original paired nominal 95 % Newcombe method 10 intervals: +18.75 [−4.28, +40.04] percentage points at locked cutoffs and −6.25 [−31.61, +19.57] at top 50. The top-50 interval conditions on observed rankings. D: detection counts at the same external K (A/B: 10/13 at 33 and 35; 14/13 at 50), each among 16 low-quality patients. K33/35 are post hoc archived-curve readouts; K50 is the original-protocol supplementary budget. No new interval or test is assigned to D.

Figure 5. Three shared screening misses in the external set. Rows show case089 (A–C), case175 (D–F), and case238 (G–I); columns show the original released RGB ultrasound image, the released reference contour (blue), and the actual initial CPU replay prediction contour (orange). Row values are CPU replay Dice against the reference and risks A and B, rather than the original archived CUDA scores. All panels preserve the full original image field and aspect ratio. In case175, the nonempty prediction includes a thin upper-edge band. Masks were captured in the six-case exact-checkpoint CPU replay; no new model inference was performed to create this figure. Original CUDA scores and primary study estimates were retained unchanged.

Figure 1

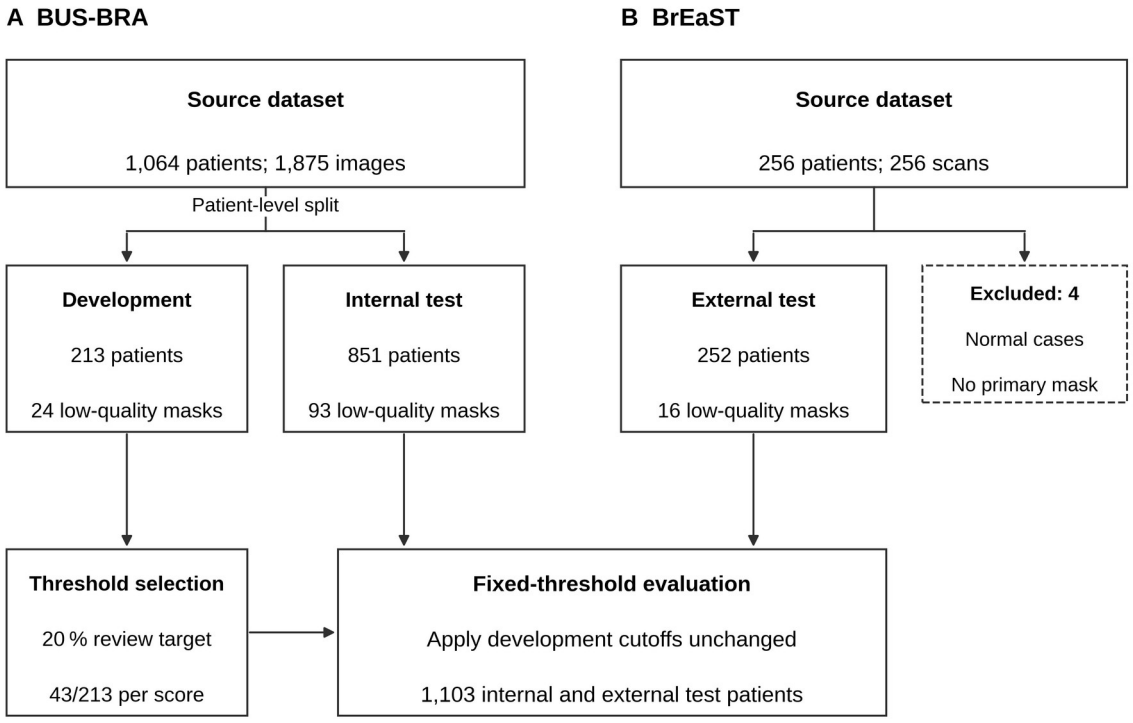


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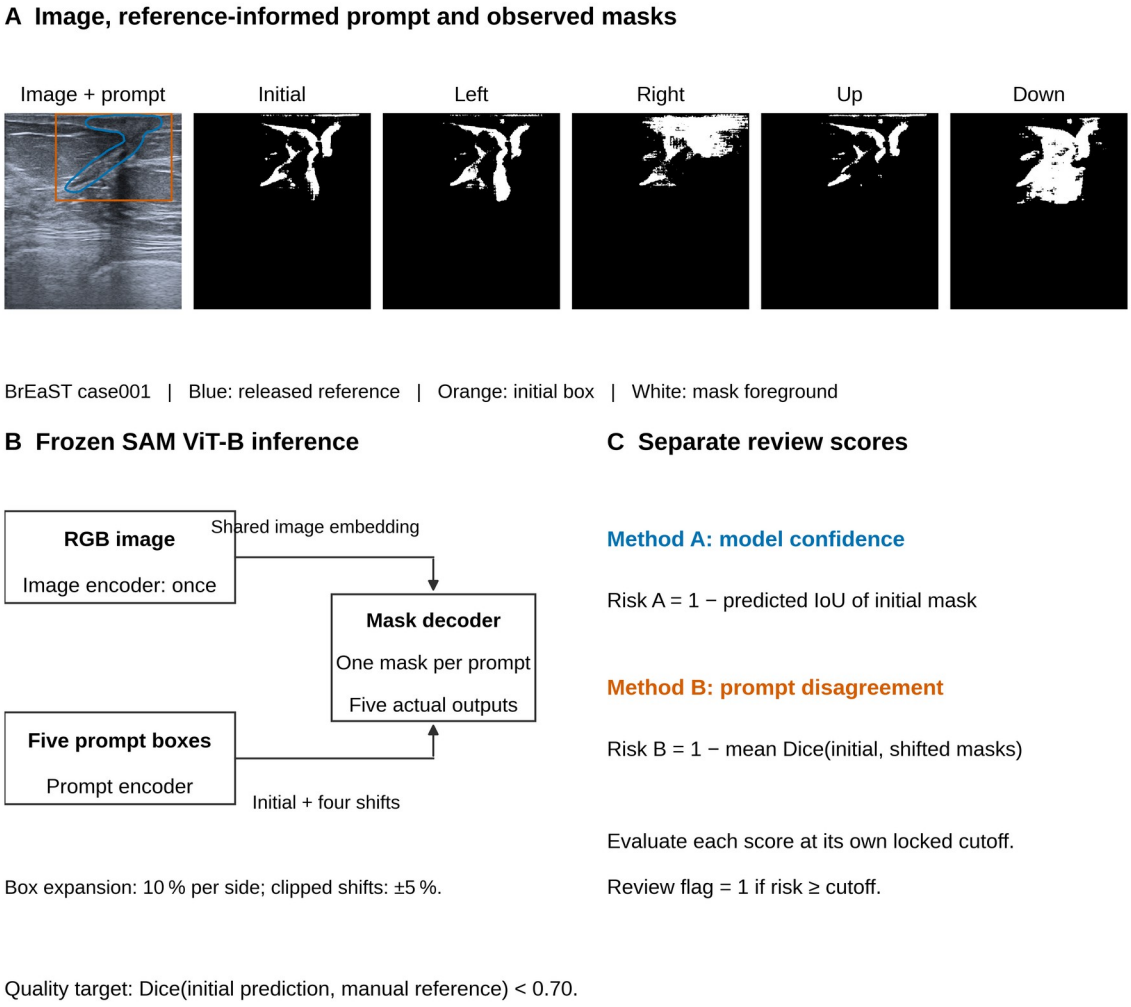


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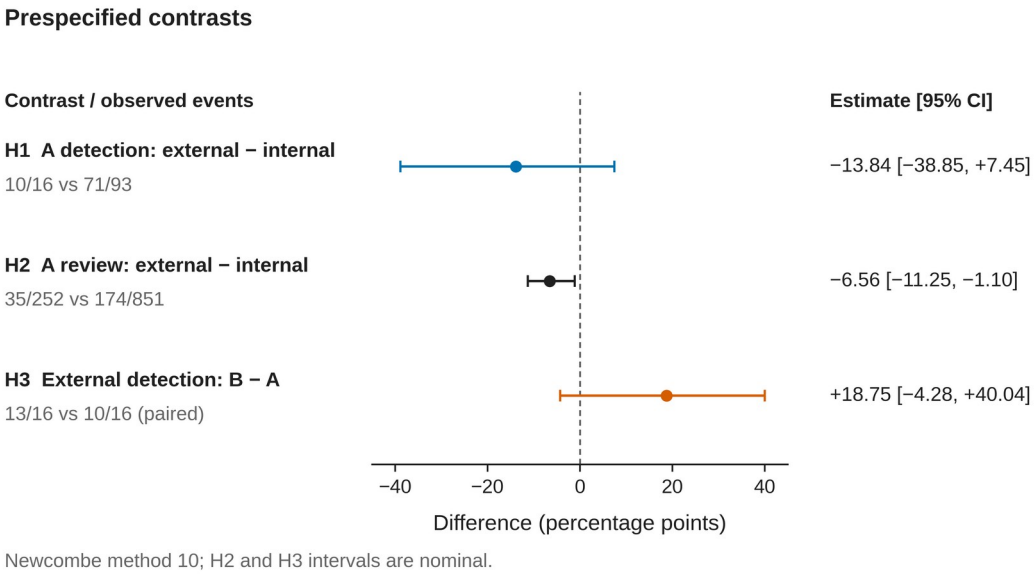


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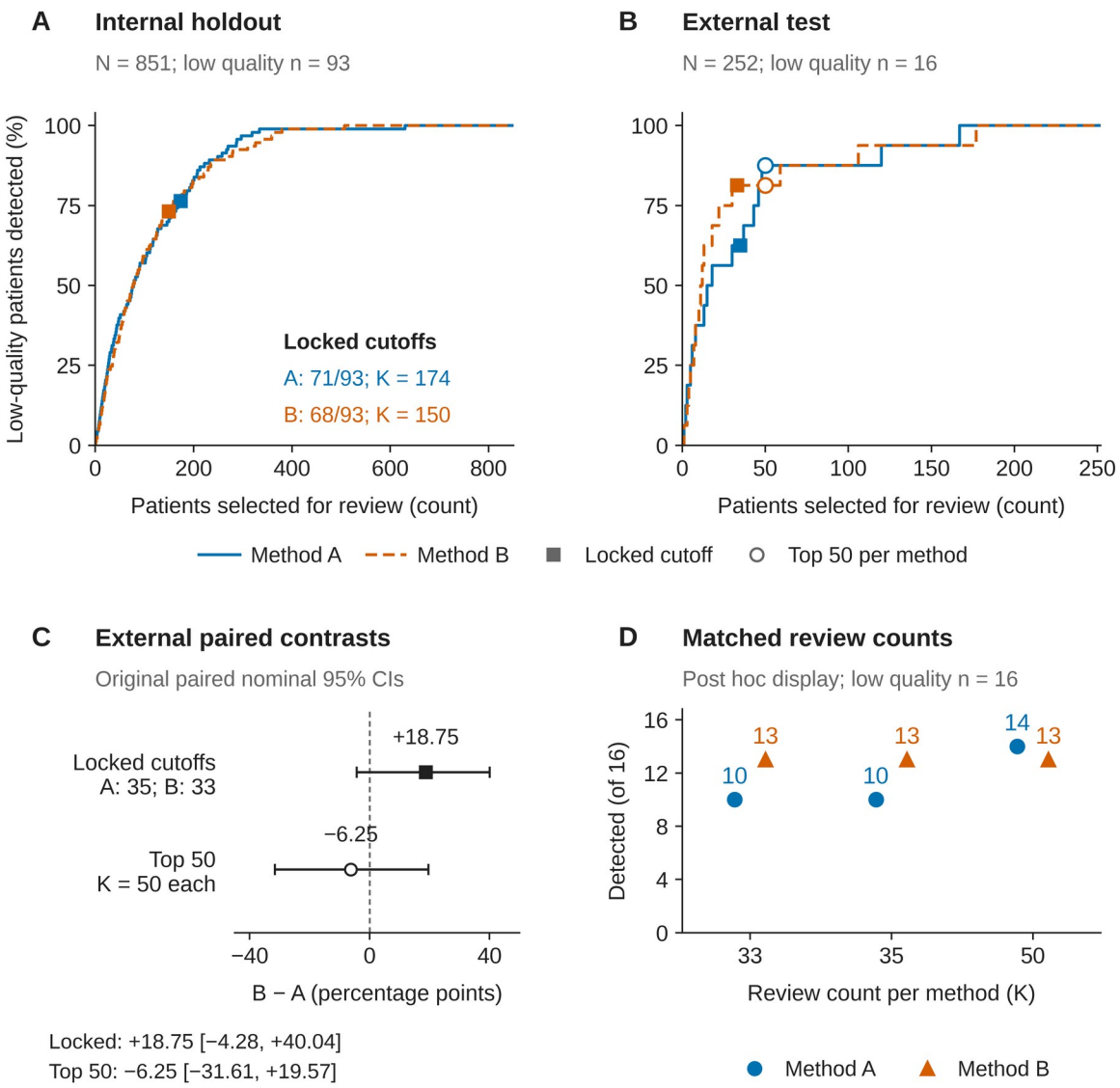


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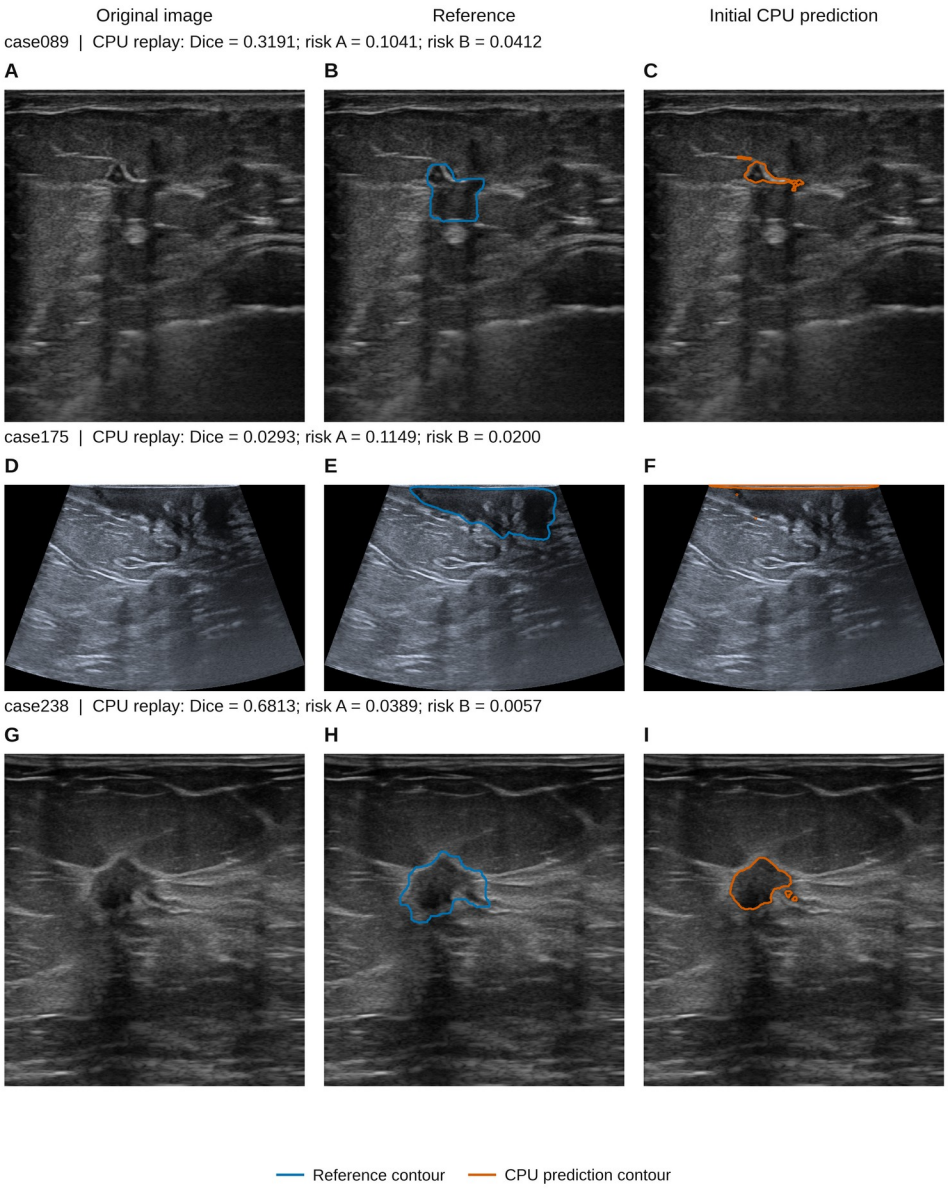


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